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Tanay Vardhan

Software Engineer, Product Designer, Aerodynamicist

EDUCATION

University of Illinois at Urbana-Champaign *B. S. Aerospace Engineering* MAY 2019 **Minor: Computer Science**

Selected Coursework:

- CS 225: Data Structures
- CS 491: Interdisciplinary Design Studio
- CS 465: Principles of User Interface Design
- CS 173: Discrete Structures
- AE 202: Aerospace Flight Mechanics
- ME 199: Automotive Design Project

ORGANIZATIONS

Illini Motorsports *Project Leader*

FORMULA SAE TEAM

- Developed Java applications to streamline configuration of computational fluid dynamics simulations on Linux cluster
- Designed Java interface to qualitatively visualize results from CFD simulations

Illinois AUV *Software Team Member*

AUTONOMOUS UNDERWATER VEHICLE TEAM

- Wrote python script to generate a heads-up display to visualize sensor data
- Used OpenCV to calculate roll, pitch and yaw angles from submarine's initial orientation

SKILLS

Full-stack Engineering

C++, PHP, Java, JavaScript, Node.js/Express.js, CSS, Python, Flask, SQL, OpenCV, Angular, Ionic

Product and Interaction Design

Photoshop, Illustrator, InDesign, InVision, Figma, Axure, Blender, Design Research, Prototyping, Wireframing,

Miscellaneous

Agile, Git, SVN, Star-CCM+ (CFD), SolidWorks (CAD), carbon-fiber layups, sanding, dremeling, tactical analysis (soccer)

WORK EXPERIENCE

Amadeus IT Group *Software Engineering Intern*

WALTHAM, MASSACHUSETTS, USA • SUMMER 2016

- Built dashboard for internal process management using PHP (Slim 3 framework) and MySQL
- Developed chrome extension for real-time notification system
- Designed interface following principles of Material Design guidelines using HTML, CSS and jQuery

Center for East Asian and Pacific Studies *Undergraduate Web Developer*

CHAMPAIGN, ILLINOIS, USA • JANUARY 2017 - PRESENT

- Redesigned 'Digital Asia', a video-streaming service for documentaries and other academic resources on East Asia
- Created modular frontend to improve performance by 20 ms per page

Athena Education *Full Stack Web Developer*

GURUGRAM, HARYANA, INDIA • SUMMER 2015

- Created Athena Tech, a web platform to automate test grading and streamline attendance processes for over 200 pre-collegiate students

RESEARCH & PROJECTS

<http://github.com/tanayv> || <http://behance.net/tanayv>

Mitigate selective exposure to news by visualizing opinion spectrum

- Research project under Prof. Wai-Tat Fu, Associate Professor of Computer Science
- Adapted open-source TensorFlow models to perform summarization and sentiment analysis to organize news articles based on opinion
- Developed classifier to determine if headline is "clickbait"
- Built python web app to visualize spectrum of contrasting articles
- Link to application: <http://opinion-spectrum.herokuapp.com>

Lapras *Best Use of Capital One's Nessie API, HackPrinceton Spring 2016*

- Web bot to analyze transaction data and make intelligent predictions on future spending with natural language processing capabilities
- Contribution: Developed backend and bot logic using PHP and MySQL

Embark *2nd Prize for Best Use of Amadeus' Innovation Sandbox*

- Flask app to generate extensive travel itineraries with minimum input
- Contribution: Created high-fidelity prototypes, designed and developed frontend using Photoshop, Illustrator, HTML, CSS, jQuery

ParkCore *Top 5 Hacks in Ford Go Detroit track at MHacks 8*

- Mobile application to crowdsource the availability of parking spots
- Contribution: Built application backend using Node.js, MySQL; designed and built frontend using HTML/CSS/JavaScript